SOF Science Unit 1

Comprehensive Topic Guide for Class III CBSE

For Ambar • June 2026

SECTION 1: PLANTS

1.1 Types of Plants (Classification)

Trees

Definition: Large plants with a thick, woody, permanent stem called trunk.
Height: Tallest among all plants (10+ meters)

Structure: Trunk → Branches → Leaves

Examples: Neem, Banyan, Mango, Coconut, Oak

Key fact: Coconut tree = "tree of life" (provides food, water, shelter, oil, fiber, leaves, wood)

Use: Timber, fruits, shade

Shrubs

Definition: Medium-sized plants with multiple woody stems, smaller than trees.
Height: 2–10 meters

Structure: Many stems from the base, bushy

Examples: Rose plant, Jasmine, Hibiscus, Guava

Characteristics: Shorter than trees, stronger than herbs

Use: Ornamental (flowers), some yield fruits

Herbs

Definition: Small, soft-stemmed plants, usually short-lived.
Height: <1 meter (dwarf)

Structure: Soft, green, non-woody stems

Examples: Mint, Tulsi, Spinach, Parsley, Coriander

Life cycle: Usually annual (live 1 year) or short-term

Use: Culinary, medicinal, spices

Climbers

Definition: Plants with weak stems that need external support to grow upward.
Mechanism: Climb up walls, poles, or trees using tendrils or twining stems

Examples: Money plant, Pea plant, Grape vine, Jasmine (some varieties)

Key feature: Weak stem + need for support = climber

Growth: Climb in an upward direction

Creepers

Definition: Plants with weak stems that spread along the ground (horizontal growth).
Mechanism: Spread on the surface; may root at joints (nodes)

Examples: Watermelon, Pumpkin, Muskmelon, Strawberry, Mint (creeping variety)

Key feature: Weak stem + spread horizontally = creeper

Growth: Grow parallel to the ground

Special trait: Often root at joints, creating new plants

KEY DISTINCTION (Common trap in exams):

Climber: Weak stem + needs support → climbs UP (vertical)

Creeper: Weak stem + NO support → spreads on GROUND (horizontal)

1.2 Parts of a Plant

Root

Function:

1. Absorbs water and minerals from soil

2. Holds/anchors the plant in soil

3. Stores food (in some plants like radish, carrot)

Types:

Tap root: One main root with smaller branches (e.g., carrot, turnip, mango)

Fibrous root: Multiple similar-sized roots (e.g., rice, wheat, grass)

Stem

Functions:

1. Holds/supports the plant upright

2. Transports water and nutrients from roots to leaves

3. Transports food made by leaves to other parts

Characteristics:

Green in many herbaceous plants (can make food)

Grows upward (positive phototropism)

Has nodes (joints where leaves attach)

Leaf

Function: Main organ for photosynthesis (making food using sunlight)

Structure:

Blade: Flat part, green

Petiole: Stalk connecting leaf to stem

Veins: Transport tubes for water and food

Key facts:

Chlorophyll: Green pigment that absorbs sunlight

Stomata: Tiny pores on leaf surface for gas exchange (CO_2 in, O_2 out)

Colour: Green because of chlorophyll

Flower

Function: Reproductive organ (produces seeds)

Parts:

Petals: Attract insects/bees for pollination

Sepals: Protect the flower when closed

Stamens: Male reproductive part (produces pollen)

Pistil: Female reproductive part

Special fact: Rafflesia has the world's largest flower (~3 ft diameter)

Fruit & Seed

Fruit Function: Protects and disperses seeds
Formation: Develops from the flower after pollination

Examples: Apple, mango, coconut, wheat grain, pea pod

Seed Function: Unit of reproduction; grows into a new plant

Parts:

Seed coat: Outer protective layer

Embryo: Baby plant inside

Food storage: Endosperm

What a seed needs to germinate:

1. Water: Activates enzymes, provides medium for growth

2. Warmth/Temperature: Optimal growth (25–30°C for most)

3. Oxygen: Needed for respiration during growth

4. Light: Mostly NOT required during germination (seeds germinate in dark soil)

1.3 Photosynthesis (Making Food)

Definition: Process by which plants make their own food using sunlight.

Simplified equation:

Water + Carbon Dioxide + Sunlight → Glucose (food) + Oxygen

(from roots) (from air) (stored) (released)

Location: Mainly in leaves (where chlorophyll is present)

Why plants need food:

Growth

Repair of damaged parts

Energy for daily activities

Key fact: Plants release oxygen as a byproduct → This oxygen is used by animals and humans for breathing.

SECTION 2: ANIMALS

2.1 Classification by Habitat

Land Animals

Live on land

Breathe using lungs

Examples: Lion, deer, cow, dog, elephant, rat, snake, lizard, bird

Water Animals

Live in water (rivers, seas, oceans)

Breathe using gills (most fish) or surface for air (whale, dolphin)

Examples: Fish, whale, dolphin, crab, octopus, starfish

Air Animals

Fly in the air

Have wings

Examples: Parrot, crow, eagle, butterfly, dragonfly, bat

Amphibians (Land + Water)

Can live both on land and in water

Lay eggs in water

Young live in water (tadpoles), adults live on land or water

Examples: Frog, toad, salamander

Key fact: Frogs are the MOST common amphibian example

2.2 Classification by Food Habit

Herbivores

Definition: Animals that eat only plants/vegetables.
Diet: Grass, leaves, fruits, seeds

Teeth: Flat molars for grinding plants

Examples: Cow, deer, goat, rabbit, sheep, elephant, zebra, giraffe

Digestive system: Long intestines to digest plants

Carnivores

Definition: Animals that eat only meat (hunt other animals).
Diet: Flesh of other animals

Teeth: Sharp, pointed (canines) for tearing meat

Examples: Lion, tiger, leopard, hawk, eagle, owl, snake

Digestive system: Short intestines (meat is easier to digest)

Omnivores

Definition: Animals that eat both plants and meat.
Diet: Vegetables, fruits, AND meat

Teeth: Mix of sharp and flat teeth

Examples: Humans, bear, pig, crow, dog, cat, monkey

Flexibility: Can survive on either plants or meat

KEY TRAPS in exams:

Cow = herbivore (NOT omnivore)

Lion = carnivore (NOT omnivore)

Human = omnivore (eats both)

2.3 Animal Body Parts and Functions

Fins (Fish)

Function: Help fish move/swim through water

Fish analogy: "Fish : Fins :: Bird : Wings"

Wings (Birds, Insects, Bats)

Function: Help animals fly through the air

Key fact: Wings are modified limbs

Hooves (Horses, Cows, Deer)

Function: Support body weight on hard ground

Protection: Hard covering

Claws/Nails

Function: Gripping, climbing, scratching, defense

Examples: Cat, tiger, eagle have sharp claws

2.4 Group Names (Collective Nouns)

Common in SOF exams:

Exam trick: "A _____ of fish" → Answer: school or pod

SECTION 3: HUMAN BODY

3.1 Sense Organs

Eyes: See light & colors → Identify objects, colors, depth

Ears: Sound waves → Communicate, detect danger

Nose: Odors → Detect good/bad smells, taste

Tongue: Flavors → Enjoy food, detect poison

Skin: Temperature, pressure → Feel texture, temperature, pain

Key facts:
Each sense has a dedicated organ

Senses are protected: eyes in sockets, ears protected by flaps

Tongue has taste buds (sensors for sweet, salty, sour, bitter)

Skin is the largest sense organ

3.2 Nutrition & Balanced Diet

Balanced diet includes:

1. Carbohydrates: Energy source
Examples: Rice, bread, wheat, potatoes, sugar

Function: Fuel for daily activities

2. Proteins: Build muscle and repair body
Examples: Eggs, meat, fish, pulses (dal), beans, milk

Function: Growth, repair, maintenance

3. Fats: Energy and hormone production
Examples: Oil, butter, nuts, seeds

Function: Energy, absorption of vitamins

4. Vitamins: Keep body healthy, boost immunity
Sources: Vegetables, fruits, eggs, milk

Types: A (vision), C (immunity), D (bones)

5. Minerals: Build bones, teeth, blood
Examples: Iron (meat, spinach), Calcium (milk, broccoli)

Function: Bone strength, oxygen transport

6. Water: Essential for all body functions
Amount: 6–8 glasses daily

Functions: Digestion, temperature control, nutrient transport

Balanced plate example:
1/4 Carbohydrates (rice/bread)

1/4 Proteins (dal/meat/egg)

1/2 Vegetables & fruits

Common deficiencies:
Only rice + bread → Lack of proteins & vitamins

Only meat → Lack of fiber

No vegetables → Lack of vitamins

3.3 Hygiene & Health

Oral Hygiene (Teeth)

Brush: Twice daily (morning & night)

Why: Remove food particles, prevent cavities, avoid decay

Duration: 2 minutes

Technique: Gentle, circular motions

Hand Hygiene

Wash hands: Before eating, after toilet, after playing

Why: Remove germs and dirt

Duration: 20 seconds with soap and water

Prevents: Diarrhea, typhoid, infections

Respiratory Hygiene

Cover mouth: When coughing or sneezing

Use: Tissue or elbow (NOT hand)

Why: Prevent spread of germs/viruses

General Hygiene

Bathe daily • Trim nails regularly • Change clothes daily • Keep surroundings clean • Drink clean water

Benefits: Better health, fewer infections, more energy

SECTION 4: ENVIRONMENT

4.1 Seasons in India

Four seasons (in order):

1. Summer (Mar–May)

Hot, dry • Crops: Wheat harvest • Activities: School holidays, heat waves • Rainfall: Minimal

2. Monsoon (Jun–Sep) [Comes after summer]

Heavy rainfall • Wet and humid • Crops: Rice, sugarcane planting • Rivers swell, floods possible

3. Autumn (Oct–Nov)

Mild temperature • Decreased rainfall • Crops: Harvesting • Weather: Pleasant

4. Winter (Dec–Feb)

Cold, dry • Frost in northern areas • Crops: Vegetables, pulses • Activities: Bonfire, festivals

Exam question: "Which season comes after summer?" → Monsoon

4.2 Water Cycle

Process:

EVAPORATION: Water from sea/lakes heats up and becomes water vapor (gas)

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CONDENSATION: Water vapor cools in upper atmosphere and forms clouds (liquid)

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PRECIPITATION: Water falls as rain (or snow in cold regions)

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COLLECTION: Water collects in rivers, lakes, seas (cycle repeats)

Key facts:
Sun's heat is the driving force

Evaporation happens from oceans (majority), lakes, rivers, soil

Rainfall is the 'collection' stage

Rainwater flows to rivers → rivers to seas → cycle repeats

4.3 Gases in the Atmosphere

Oxygen (O_2)

Source: Released by plants during photosynthesis

Use: Humans and animals breathe to get energy from food

Percentage: ~21% of air

Importance: Essential for life

Carbon Dioxide (CO_2)

Source: Exhaled by animals/humans during respiration

Use: Plants absorb CO_2 for photosynthesis (food-making)

Percentage: ~0.04% of air

Cycle: Tight loop between plants and animals

Nitrogen (N_2)

Percentage: ~78% of air

Use: Limited direct use (bacteria fix nitrogen for plants)

Importance: Nutrient for plant growth

Interconnection:

Plants (photosynthesis) → Release O_2 → Used by animals

Animals (respiration) → Release CO_2 → Used by plants

Balance maintained by nature

Exam questions:

"Which gas do plants need?" → Carbon dioxide (CO_2)

"Which gas do we breathe in?" → Oxygen (O_2)

4.4 Light & Shadows

Shadow formation:

Requirement: Light source + Opaque object + Screen

Mechanism: Object blocks light rays

Properties:

- Dark area on the opposite side of light source

- Shape matches the object

- Length changes with sun position (long in morning/evening, short at noon)

4.5 Forces: Push & Pull

Definition: Push and pull are types of forces (actions that cause movement or change).

Push

Definition: Force applied to move object AWAY from you

Examples: Push a door, push a cart, kick a ball, punch a bag

Pull

Definition: Force applied to move object TOWARD you

Examples: Pull a door, pull a rope, draw a bucket from well, pull hair

EXAM PREPARATION STRATEGY

Common Traps to Avoid

Climber grows on ground → NO: Climber climbs UP; creeper spreads on GROUND

Cow is omnivore → NO: Cow eats only plants (herbivore)

Frogs live only in water → NO: Frogs are amphibians (land + water)

Light is NOT needed for seed germination → Light often NOT needed; water, warmth, oxygen are essential

Spring comes after summer → NO: Monsoon comes after summer (in India)

Fish release oxygen → NO: Plants release oxygen; fish use oxygen

Only carbs give energy → Carbs + fats give energy; proteins help growth

High-frequency SOF Question Types

Type 1: Definition-based

"A _____ is a plant with a strong stem that is very tall." → Answer: Tree

Type 2: Example-based

"Which of these is a herb?" → Answer: Mint / Tulsi / Coriander

Type 3: Analogy-based

"Fish : Fins :: Bird : ____" → Answer: Wings

Type 4: Classification

"Deer eats grass. What kind of animal is a deer?" → Answer: Herbivore

Type 5: Function-based

"Which part of the plant absorbs water?" → Answer: Root

Type 6: Cause-effect

"If a plant loses all its leaves, what happens?" → Answer: Plant cannot make food and will die

Type 7: Application-based

"Mohan eats only rice. What should he add to his diet?" → Answer: Proteins and vegetables

KEY FACTS CHECKLIST FOR AMBAR

☐ Climber vs Creeper distinction

☐ Four types of plants (tree, shrub, herb) + Climber + Creeper

☐ Parts of plant: Root, Stem, Leaf, Flower, Fruit, Seed

☐ Photosynthesis: Sunlight + Water + CO_2 → Food + Oxygen

☐ Chlorophyll = green pigment

☐ Rafflesia = largest flower

☐ Coconut tree = tree of life

☐ Sunflower follows the sun

☐ Herbivore, Carnivore, Omnivore definitions + examples

☐ Frog = amphibian (both water and land)

☐ Fish group = school; Birds group = flock

☐ Five sense organs and their functions

☐ Brush teeth TWICE daily

☐ Balanced diet = carbs + proteins + fats + vitamins + minerals + water

☐ Monsoon comes after summer (in India)

☐ Water cycle: Evaporation → Condensation → Precipitation

☐ Plants release O_2; animals/humans need O_2

☐ Shadow = light blocked by object

☐ Push = force away; Pull = force toward

QUICK REVISION DRILL

Answer these 10 questions in 5 minutes:

1. What is the green pigment in leaves called?

2. Name a creeper plant.

3. Which animal eats only plants?

4. Frogs can live on _____ and in _____.

5. How many times should you brush teeth daily?

6. The _____ season comes after summer.

7. Water enters the plant through the _____.

8. A group of fish is called a _____.

9. Which sense organ helps you smell?

10. Photosynthesis happens mainly in the _____.

Answers:

1. Chlorophyll

2. Watermelon / Pumpkin / Mint

3. Herbivore (e.g., deer, cow)

4. Land, Water

5. Twice

6. Monsoon

7. Root

8. School

9. Nose

10. Leaf

Good luck with SOF Science Unit 1, Ambar!